Syllabus

ELECTROSTATICS

- Force and electric field due to continuous charge distribution
- Field lines – Flux
- Gauss law – and its Applications
- Electric potential – due to continuous charge distribution – equipotential line/ surface
- Poisson's equation and its solution (Limit to 1-D).
- Workdone in assembling a charge distribution
- Conductors and Capacitors
- Boundary conditions for conductors
- Electric dipole – field lines – Force and torque on a dipole due to external static electric field – Interaction energy between two dipoles.
- Multipole expansion - Upto Quadrupole only
- Polarization – restricting to linear, isotropic and homogeneous medium – surface and volume bound charge density
- Electric displacement vector – Dielectric permittivity and susceptibility – Gauss law in materials
- Workdone in assembling free charges in the vicinity of dielectric material
- Complete boundary conditions

Syllabus

MAGNETOSTATICS

- Force law – line current, surface current and volume current densities
- Biot-Savart Law – Properties of B

- Magnetic flux – Div B – Magnetic Vector Potential A – Information about

  magnetic scalar potential A
- Curl B – Ampere's law
- Properties of A – Multipole expansion – magnetic dipole moment – Force and torque on a magnetic dipole due to external static magnetic field.
- Magnetization – Bound volume and surface current densities
- Auxillary field H – Magnetic susceptibility and permeability – general information about para, dia and ferro magnetism
- Boundary conditions
- Force on charged particle under electric and magnetic fields

Syllabus

ELECTROMAGNETICS

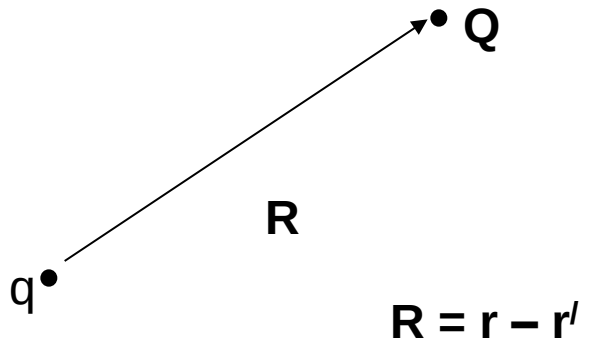
- Electromotive force – Faraday's law of electromagnetic induction
- Self and mutual inductance
- Displacement current density
- Maxwell's equation for electromagnetic field
- Derivation of wave equation – Plane wave solution – Poynting vector S.
- Reflection and transmission of electromagnetic radiation at boundary

SUGGESTED BOOKS FOR REFERENCE

- Introduction to Electrodynamics by Griffiths
- Electromagnetic fields by Wangsness
- Electricity and Magnetism by Mahajan and Rangwala
- Elements of electromagnetics by Sadiku
- Fields and wave electromagnetics by Cheng
- Electromagnetics by Kraus

Electrostatics

Coulomb's law



Force on a test charge Q due to a point charge q

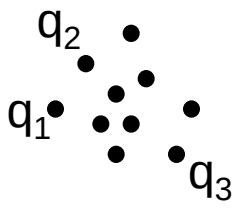
given by Coulomb's law (based on experiments)

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{R^2}\hat{e}_R = Q\vec{E}$$

ε_0 **is the permittivity of free space**

$$\varepsilon_0 = 8.85 \times 10^{-12}\ \frac{C^2}{N.m^2}$$

Coulomb's law and superposition principle forms the physical input for electrostatics



Source charges



Test Charge

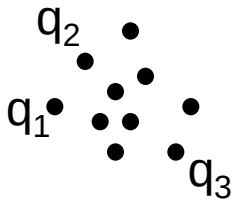
Principle of superposition states that the interaction between any two charges is completely unaffected by the presence of others

$\mathbf{F_1}$ force on Q due to q_1, $\mathbf{F_2}$ force on Q due to q_2…..

Vector sum of all the individual forces

$$\mathbf{F} = \mathbf{F_1} + \mathbf{F_2} + \mathbf{F_3} + \ldots\ldots..$$

The Electric field



Q

Test Charge

Source charges

Total force on Q is

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + ...$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 Q}{R_1^{\,2}}\hat{e}_{R_1} + \frac{q_2 Q}{R_2^{\,2}}\hat{e}_{R_2} + ...\right)$$

$$= \frac{Q}{4\pi\varepsilon_0}\left(\frac{q_1}{R_1^{\,2}}\hat{e}_{R_1} + \frac{q_2}{R_2^{\,2}}\hat{e}_{R_2} + ...\right)$$

$$\vec{F} = Q\vec{E}$$

Notations used in the course

i^{th} Source charge	q_i
Test charge	Q
Position vector of Q *or* the point of interest	$\vec{r}$
Position vector of q_i	$\overrightarrow{r_i}'$
Vector from q to Q	$\vec{r}_{qQ}$

Coulomb's law – *Experimental observation*

Force on a test charge Q due to a point charge q located at the origin

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}\hat{r}$$

ε_0 - permittivity of free space

$\varepsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{N.m^2}$

Independent of the coordinate system

When there is a collection of discrete charges

$$\vec{F} = \sum_{i=1}^{n}\vec{F}_i = \frac{1}{4\pi\varepsilon_0}\left(\sum_{i=1}^{n}\frac{q_i}{r_{q_iQ}^2}\hat{r}_{q_iQ}\right)Q$$

by the principle of superposition

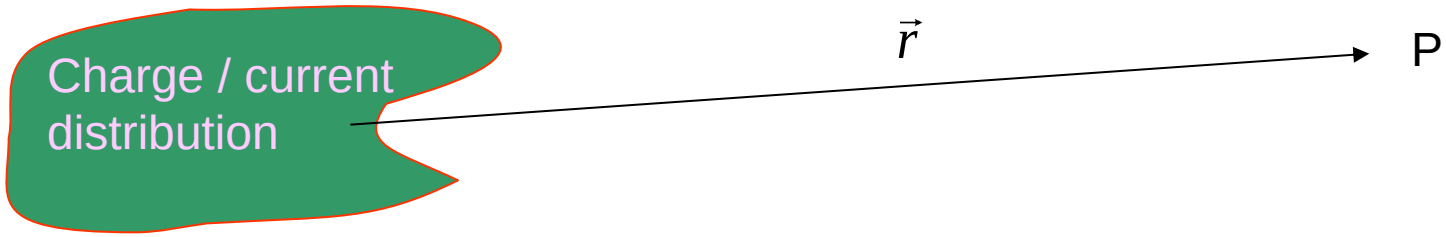
Electric field $\vec{E} = \vec{F}/Q$

$$= \frac{1}{4\pi\varepsilon_0}\left(\sum_{i=1}^{n}\frac{q_i}{r_{q_iQ}^2}\hat{r}_{q_iQ}\right)$$

Now, classical electrodynamics deals with electric and magnetic fields and interactions due to the *macroscopic* distributions of charges and currents

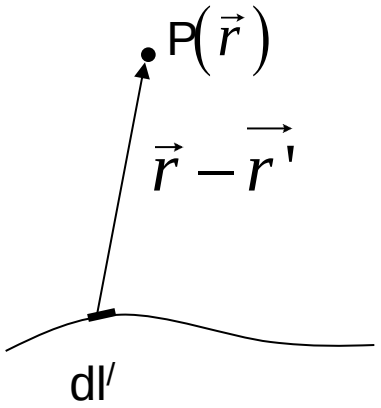
The electric and magnetic FILEDS due to a collection of charges/currents, at a point away from the charges/currents will be our aim in this course.

The concepts of localized charges and currents assume the validity of certain mathematical limiting processes wherein, the charge and current distributions are assumed to be localized on infinitesimal volumes.



Continuous charge distributions

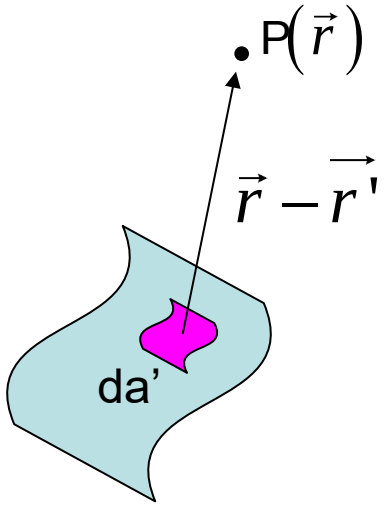
Line charge



Line charge, λ

$$\vec{E}=\frac{1}{4\pi\varepsilon_0}\int_c \frac{\lambda dl'}{\left|\vec{r}-\vec{r}'\right|^3}\left(\vec{r}-\vec{r}'\right)$$

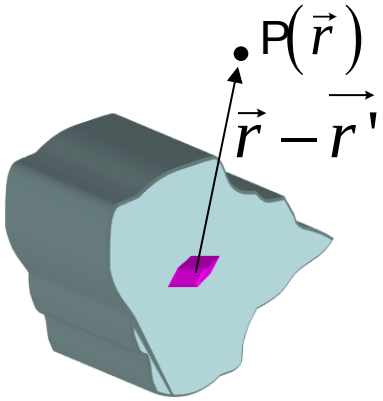
Surface charge



Surface charge, σ

$$\vec{E}=\frac{1}{4\pi\varepsilon_0}\int_s \frac{\sigma da'}{\left|\vec{r}-\vec{r}'\right|^3}\left(\vec{r}-\vec{r}'\right)$$

Volume charge

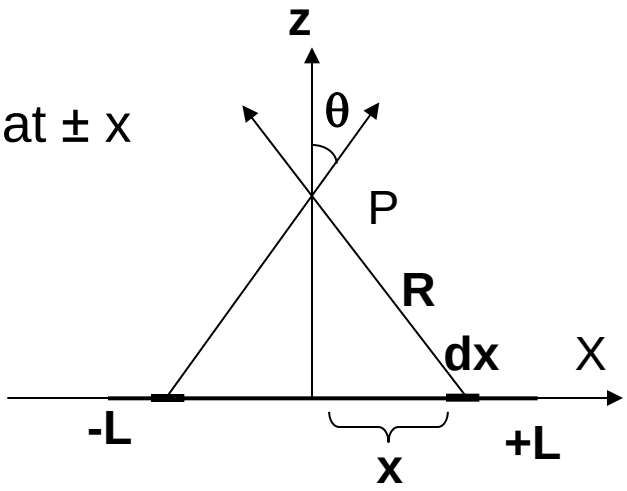


Volume charge, ρ

$$\vec{E}=\frac{1}{4\pi\varepsilon_0}\int_v \frac{\rho dV'}{\left|\vec{r}-\vec{r}'\right|^3}\left(\vec{r}-\vec{r}'\right)$$

Example

Find the electric field a distance z above the mid point of a straight line segment of length 2L, which carries a uniform line charge λ



Two symmetrically placed pairs of length dx at ± x

horizontal components cancel !!!

net field of the pair is

$$d\vec{E} = 2\frac{1}{4\pi\varepsilon_0}\left(\frac{\lambda dx}{R^2}\right)\cos\theta\hat{e}_z$$

$$\cos\theta = \frac{z}{R} and$$

$$R = \sqrt{z^2 + x^2}$$

The limits of x are 0 and L

$$E = \frac{1}{4\pi\varepsilon_0}\int_0^L \frac{2\lambda z}{\left(z^2 + x^2\right)^{3/2}}dx$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{2\lambda L}{z\sqrt{z^2 + L^2}}$$

It is in the $\hat{e}_z$ direction

For points far from line (z >> L) $\sqrt{z^2 + L^2} = \sqrt{z^2}$

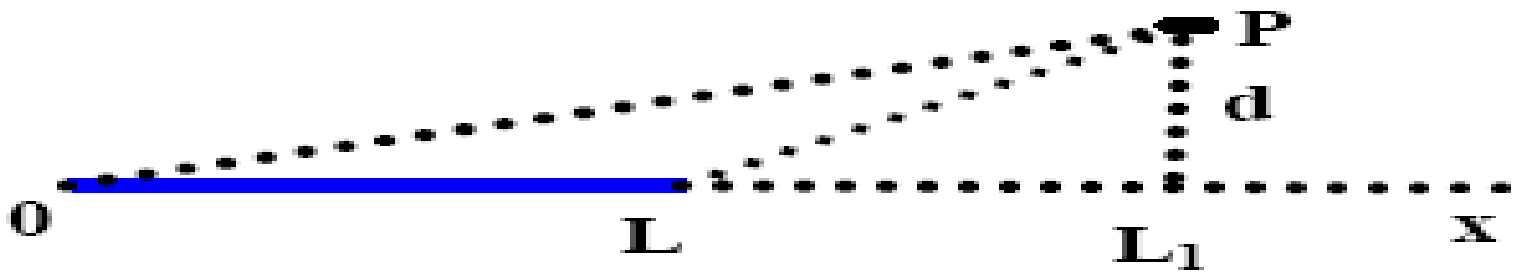
$$E \cong \frac{1}{4\pi\varepsilon_0}\frac{2\lambda L}{z^2}$$

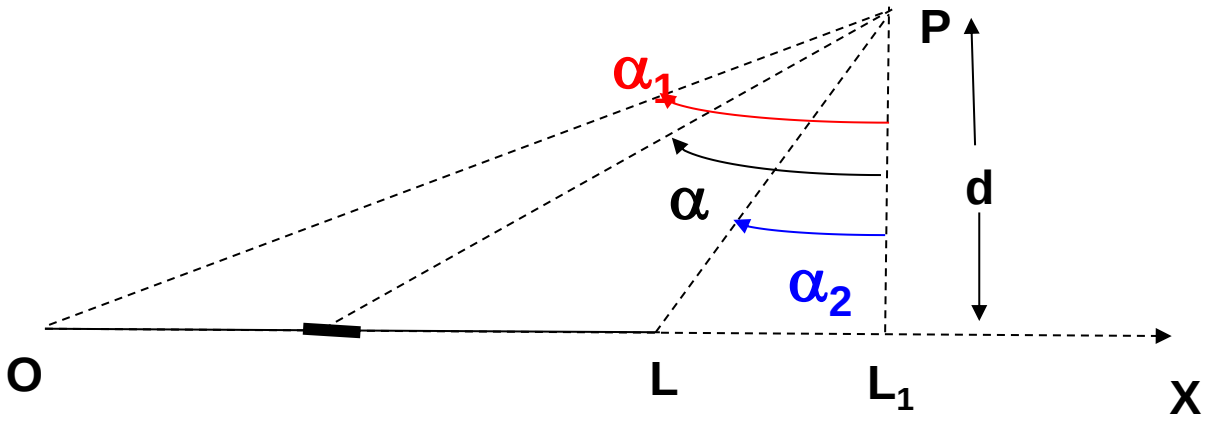
In the limit L $\rightarrow \infty$ Z = L $E = \frac{1}{4\pi\varepsilon_0}\frac{2\lambda}{z}$

EXAMPLES

Evaluate force on a test charge at P.

a) Line charge of length L and uniform line charge density λ_0.





$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int_0^L \frac{\lambda_0 dx\left[\left(L_1 - x\right)\hat{e}_x + d\hat{e}_y\right]}{\left[\left(L_1 - x\right)^2 + d^2\right]^{3/2}}$$

$$L_1 - x = d\tan\alpha \qquad dx = -d\sec^2\alpha d\alpha$$

$$\left[\left(L_1 - x\right)^2 + d^2\right]^{3/2} = d^3\sec^3\alpha$$

$$\vec{E} = \frac{-\lambda_0}{4\pi\varepsilon_0 d}\int_{\alpha_1}^{\alpha_2}\left[\left(\sin\alpha d\alpha \hat{e}_x\right)+\left(\cos\alpha d\alpha\right)\hat{e}_y\right]$$

$$= \frac{-\lambda_0}{4\pi\varepsilon_0}\frac{1}{d}\left[\left(\cos\alpha_1-\cos\alpha_2\right)\hat{e}_x+\left(\sin\alpha_2-\sin\alpha_1\right)\hat{e}_y\right]$$

$$\cos\alpha_1 = \frac{d}{\sqrt{L_1{}^2+d^2}} \qquad \cos\alpha_2 = \frac{d}{\sqrt{\left(L_1-L\right)^2+d^2}}$$

$$\sin\alpha_1 = \frac{L_1}{\sqrt{L_1{}^2+d^2}} \qquad \sin\alpha_2 = \frac{L_1-L}{\sqrt{\left(L_1-L\right)^2+d^2}}$$

NOTE:

(a) If P is such that the coordinate is $(\frac{L}{2}, d)$ then $\alpha_1 = \alpha_2$. Therefore,

$$\vec{E} = \frac{+\lambda_0}{4\pi\epsilon_0} \cdot \frac{1}{d} \left[2 \cdot \sin\alpha_1 \, \hat{e}_y \right]$$

(b) For a semi-infinite line,

$\alpha_1 = \frac{\pi}{2}$ and $\alpha_2 = 0$.

$\cos\alpha_1 = 0 \quad \sin\alpha_1 = 1$

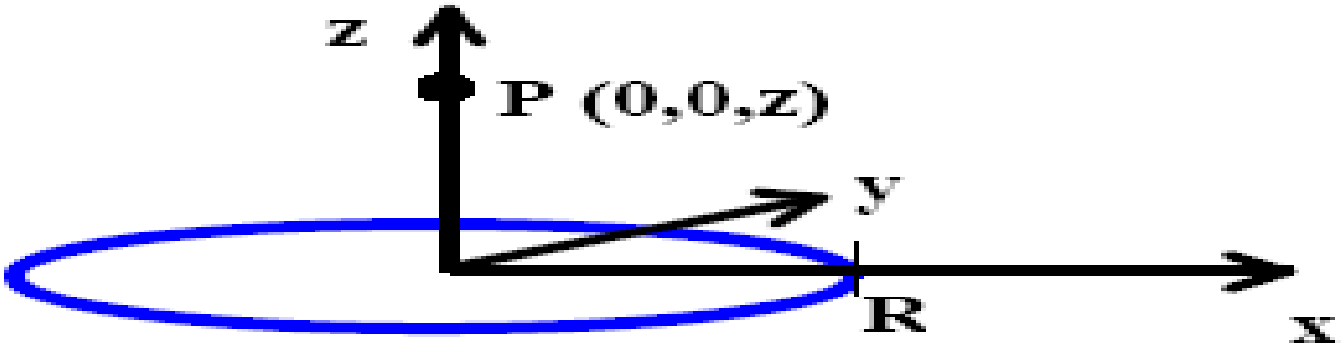
$\cos\alpha_2 = 1 \quad \sin\alpha_2 = 0$



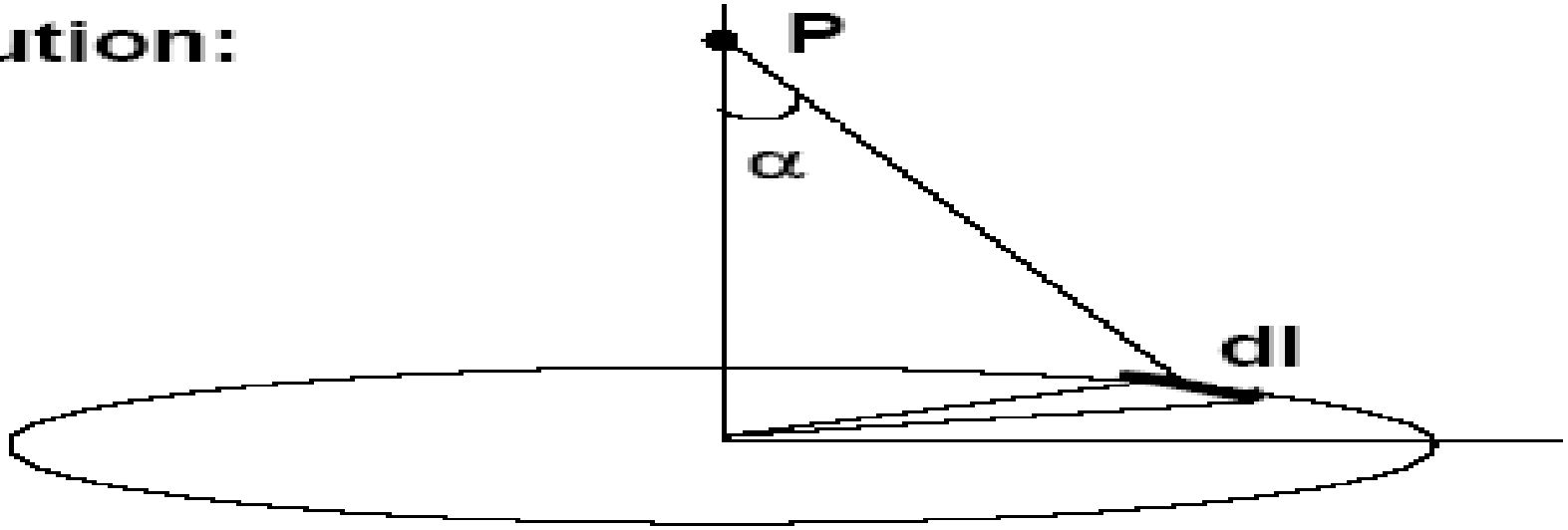
$$\vec{E} = \frac{-\lambda_0}{4\pi\epsilon_0} \cdot \frac{1}{d} \left[-\hat{e}_x - \hat{e}_y \right]$$

$$\vec{E} = \frac{\lambda_0}{4\pi\epsilon_0 d} \left[\hat{e}_x + \hat{e}_y \right]$$

2) A circular line charge of radius R and uniform line charge density λ_0.



Solution:



$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int\left(\frac{\lambda_0 dl\left(z\hat{e}_z - \mathrm{R}\hat{\mathrm{e}}_\rho\right)}{\left(z^2+R^2\right)^{3/2}}\right)$$

Here dl is $Rd\varphi$ and integration is performed over $\varphi = 0$ to 2π. As the integration,

$$\oint_{\varphi} \mathrm{R}\, d\varphi\, \hat{\mathrm{e}}_\rho = 0$$

The line integration is for $d\varphi$.

Therefore,

$$\vec{E} = \frac{\lambda_0 R z}{4\pi\varepsilon_0 \left(z^2 + R^2\right)^{3/2}} \oint d\varphi \hat{e}_z$$

$$\vec{E} = \frac{\lambda_0 R z}{2\varepsilon_0 \left(z^2 + R^2\right)^{3/2}} \hat{e}_z$$

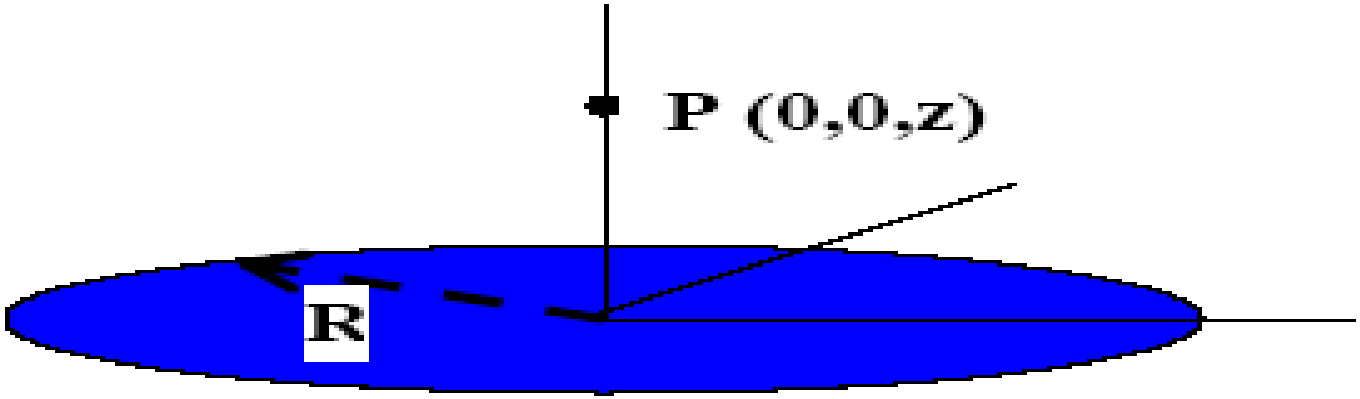
For z = 0, $E_z = 0$.

For very large values of z (z>>R),

$$\vec{E} = \frac{2\pi R \lambda_0}{4\pi\varepsilon_0 z^2} \hat{e}_z = \frac{q}{4\pi\varepsilon_0 z^2} \hat{e}_z$$

The line element is considered as a point source of charge *q*.

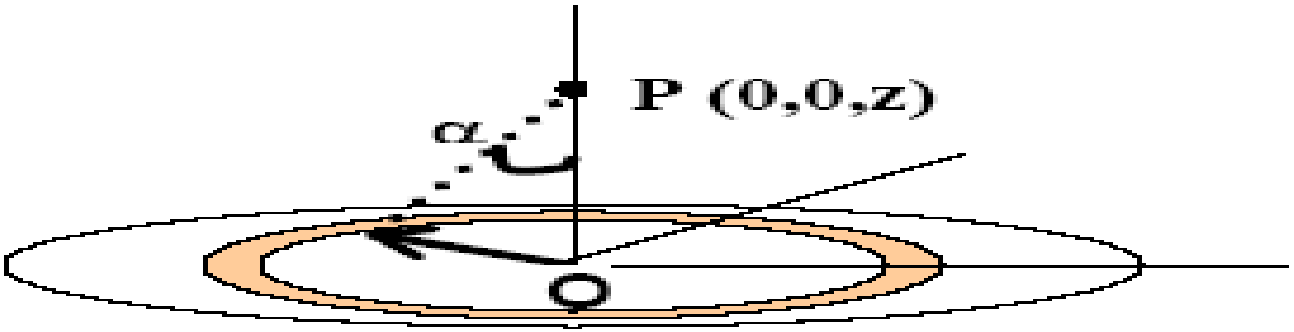
3. A circular sheet of charge with uniform surface charge density σ_0.



If several concentric circular line charges, with radii ranging from 0 to R, are closely stacked, we get the sheet of charge of radius R. In this case,

$\lambda_0 = \sigma_0 . d\rho$: $\quad R \rightarrow \rho$

$$\vec{E} = \int_{\rho=0}^{R} \frac{\sigma_0 z \rho d\rho}{2\varepsilon_0 \left(z^2 + \rho^2\right)^{3/2}} \hat{e}_z$$



Using the transformation $\rho = z \tan\varphi$, the above expression gets transformed into:

$$\vec{E} = \frac{\sigma_0 z}{2\varepsilon_0} \int_{\varphi=0}^{\alpha} \frac{z \tan\varphi z \sec^2\varphi d\varphi}{z^3 \sec^3\varphi} \hat{e}_z$$

$$\vec{E} = \frac{\sigma_0}{2\varepsilon_0}(1-\cos\alpha)\hat{e}_z$$

It is also possible to start from the basics by taking a small area element $\rho\, d\rho\, d\varphi$ and integrating for φ and ρ.

Note:

If the circular sheet extends to infinity, ie. $R \rightarrow \infty$, then $\alpha = \pi/2$ and therefore, $|E| = \sigma_0/2\varepsilon_0$.

Electrostatic Potential

Electrostatic field is conservative vector field as it is derived from central force; all central forces are conservative forces. Vice-versa is not true, i.e., all conservative field need not be central force.

A conservative field can be expressed as the gradient of a scalar potential. The electrostatic potential follows as

$$\vec{E}(\vec{r}) = -\vec{\nabla}\phi(\vec{r}) \quad \Leftrightarrow \vec{\nabla} \times \vec{E}(\vec{r}) = 0$$

where $\phi(\vec{r})$ is electrostatic potential

Electrostatic potential is irrotational in the region in which it can be defined

Electrostatic potential for a single point charge can be derived from electrostatic field as

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \frac{q}{\left|\vec{r} - \vec{r}'\right|} + C$$

Electrostatic Potential

A system of point charges q_i located at the points $\vec{r}_i'$

$$\phi(\vec{r}) = \sum_{i=1}^{n} \frac{1}{4\pi\varepsilon_0} \frac{q_i}{\left|\vec{r} - \vec{r}_i'\right|} + C$$

For Line charge, λ

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \int_C \frac{\lambda dl'}{\left|\vec{r} - \vec{r}'\right|} + C$$

For surface charge, σ

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \int_s \frac{\sigma da'}{\left|\vec{r} - \vec{r}'\right|} + C$$

For Volume charge, ρ

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \int_v \frac{\rho dV'}{\left|\vec{r} - \vec{r}'\right|} + C$$

The Dirac delta Function

- Consider the vector function $V = \frac{1}{r^2}\hat{r}$

At every location, V is directed radially outwards.
Taking divergence

$$\nabla \bullet V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{1}{r^2}\right) = \frac{1}{r^2}\frac{\partial}{\partial r}(1) = 0$$

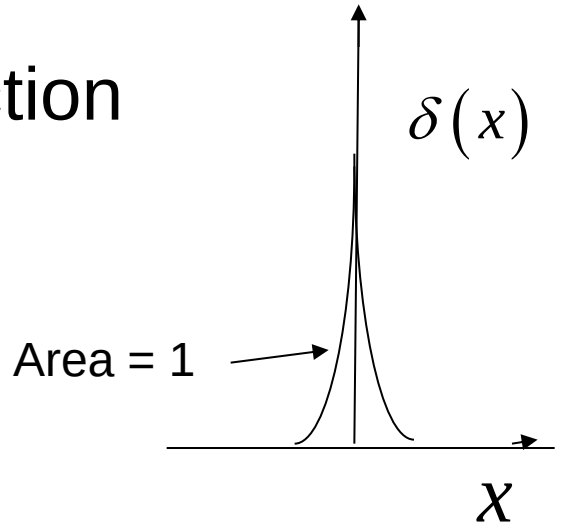
But, when r = 0, V blows up, because r is in the denominator. Thus $\nabla \bullet V$ has a special property that it vanishes **except** at the origin.

Similarity is the case of the density (mass per unit volume) of a point particle.

This is a peculiar function called as Dirac delta Function.

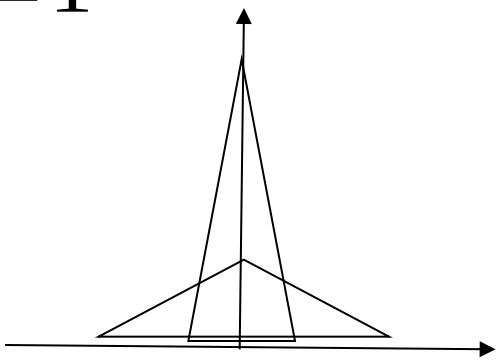
One Dimensional Dirac Delta Function

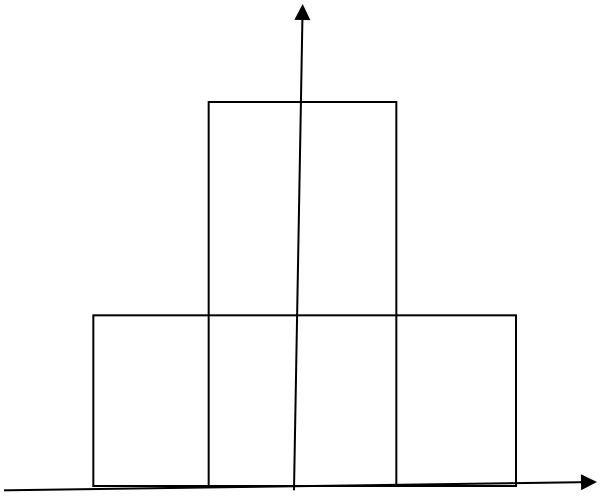
This an infinitely high,
infinitesmally narrow spike with area = 1



$$\delta(x) = \begin{cases} 0 & \textbf{If } x \neq 0 \\ \infty & \textbf{If } x = 0 \end{cases}$$

and $$\int_{-\infty}^{\infty} \delta(x)\,dx = 1$$





Area = 1

- If f(x) is some function, then

  f(x) δ(x) becomes zero except at x = 0

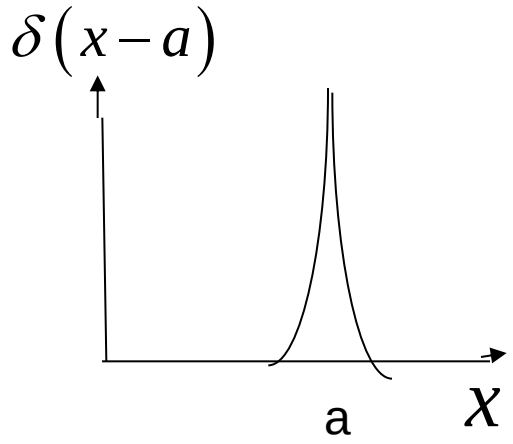
  We may write,

$$\int_{-\infty}^{\infty} f(x)\,\delta(x)\,dx = f(0)\int_{-\infty}^{\infty} \delta(x)\,dx = f(0)$$

Of course, we can shift the spike from x = 0 to x=a

$$\delta(x-a) = \begin{cases} 0 & \textbf{If } x \neq a \\ \infty & \textbf{If } x = a \end{cases}$$

$$\int_{-\infty}^{\infty} f(x)\,\delta(x-a)\,dx = f(a)$$



Three Dimensional Delta Function

The generalized 3-D Delta function is

$$\delta^3\left(\vec{r}\right) = \delta(x)\delta(y)\delta(z)$$

where $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

is the position vector from origin to (x,y,z)

Volume Integral $$\int_{-\infty}^{\infty} \delta(x)\delta(y)\delta(z)\,dxdxdz = 1$$

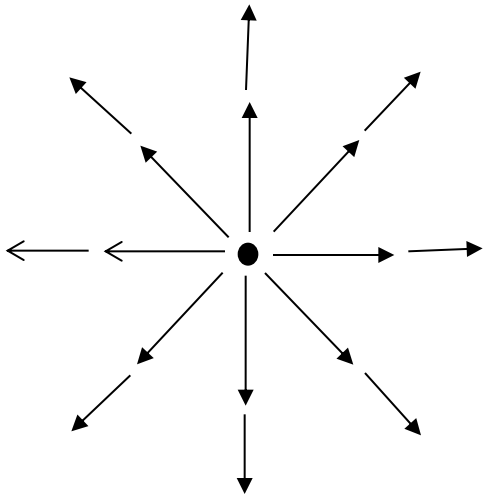
Generalizing $$\int_{-\infty}^{\infty} f(\vec{r})\,\delta^3(r-a)\,d\tau = f(a)$$

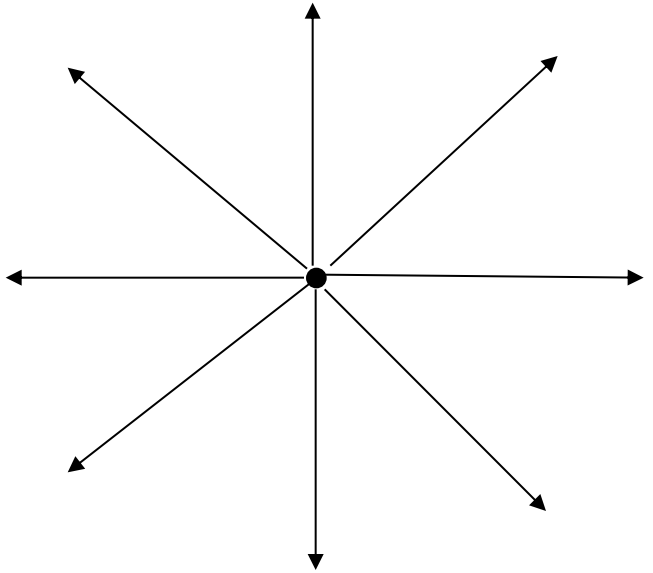
Field lines, Flux and Gauss's law

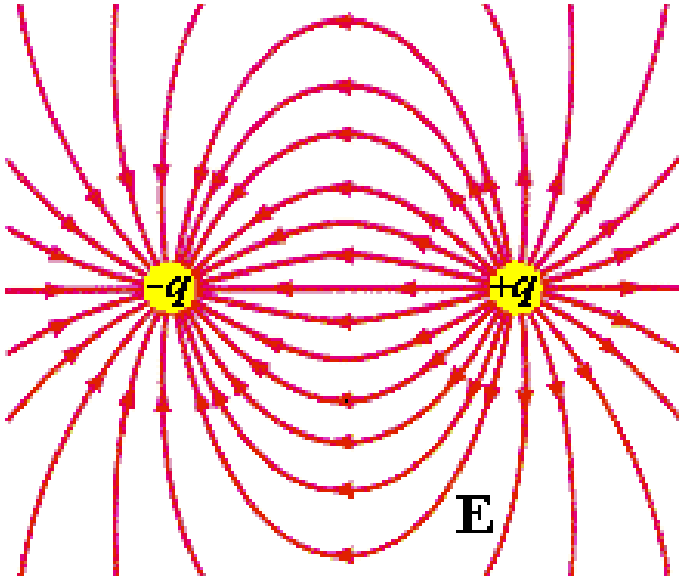
Field Lines

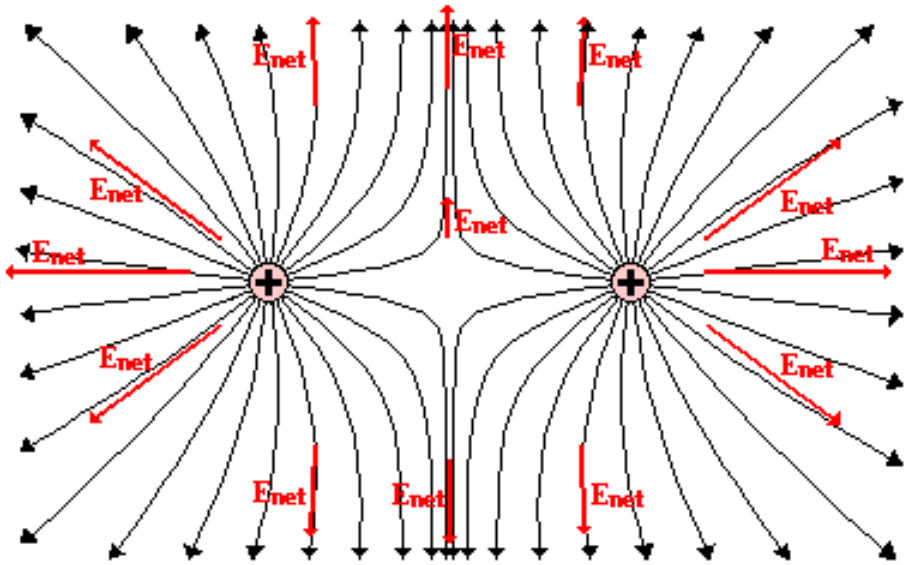
$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{e}_r \quad ----(1)$$

Field lines falls like $1/r^2$

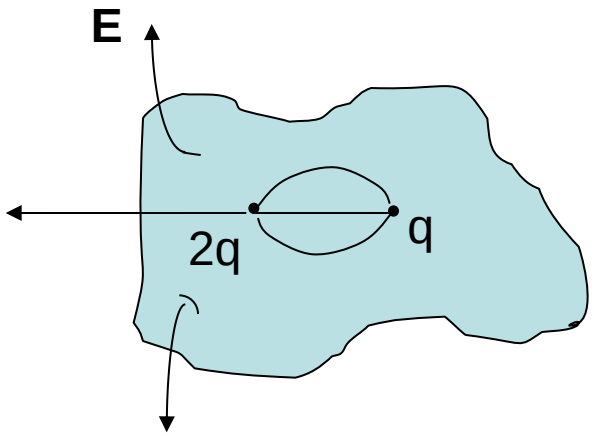




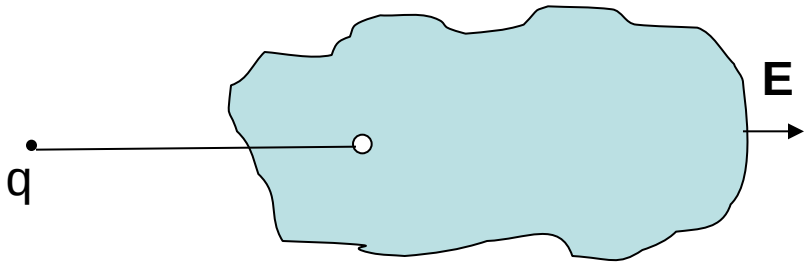




Field lines never crosses each other

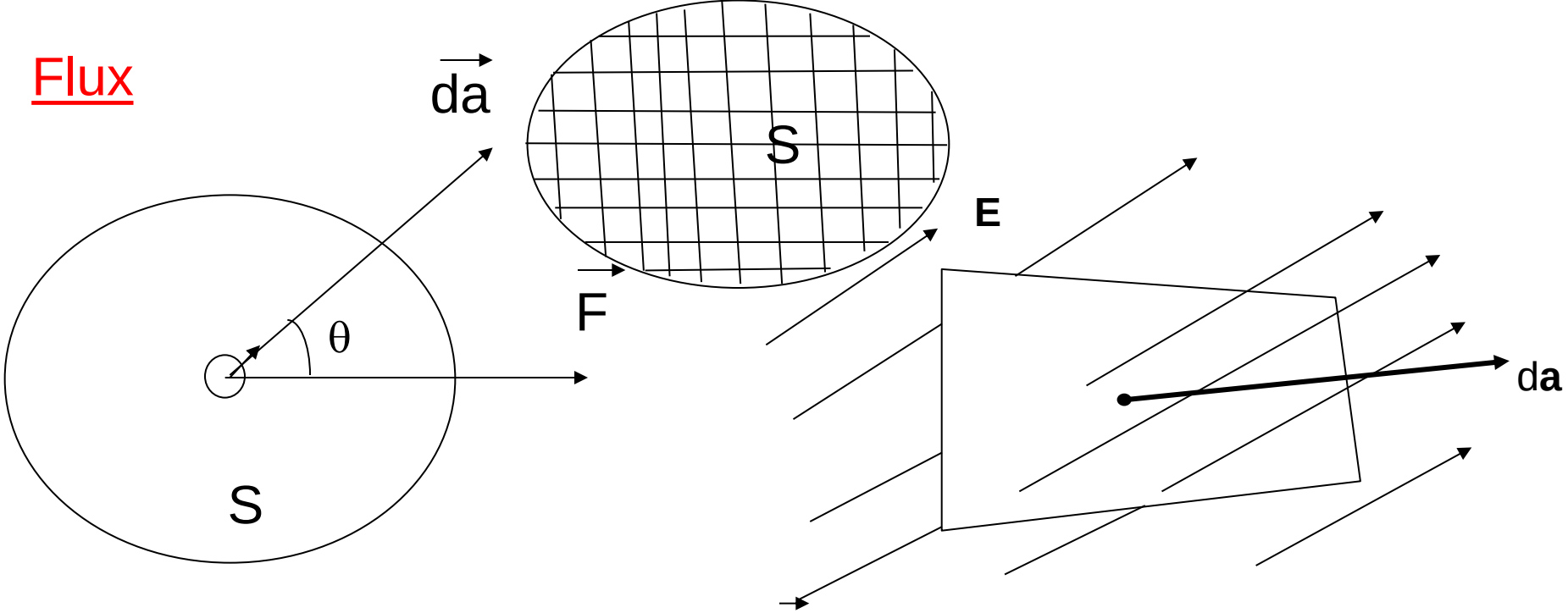


Field lines that originate on a positive charge must either pass out through the Surface or else terminate on a negative charge inside



A charge outside the surface will contribute nothing to the Flux, since its field lines pass in one side and come out through the other side

Flux



At each elemental area evaluate $\vec{F}$, multiply its component in the direction of $\vec{da}$ by the magnitude of $\vec{da}$ and add all the quantities

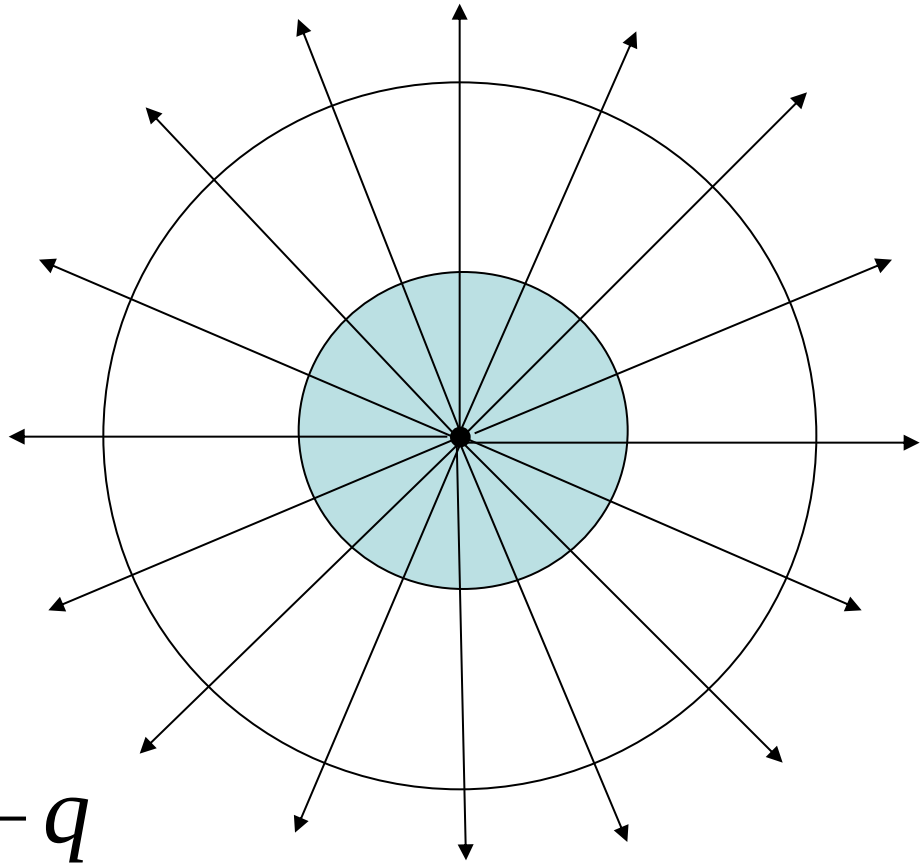
Surface integral of $\vec{F}$ over S

$$\int_s F\cos\theta da = \int_S \vec{F}.\hat{n}da = \int_s \vec{F}.d\vec{a}$$ = FLUX

Only the area in the plane perpendicular to **F**

For a point charge q at the origin, the flux of $\vec{E}$ through a sphere of radius r is

$$\oint \vec{E} \cdot d\vec{A} = \int \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2}\hat{e}_r\right)\cdot\left(r^2 \sin\theta d\theta d\phi \hat{e}_r\right) = \frac{1}{\varepsilon_0} q$$



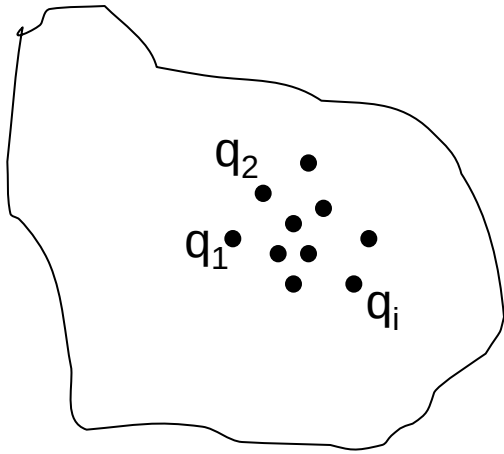
True for any odd surface

Flux through any surface enclosing charge is $= \frac{1}{\varepsilon_0} q$

Suppose we have a bunch of charges

Superposition principle to be employed

$$\vec{E} = \sum_{i=1}^{n} \vec{E}_i$$



Flux through a surface that enclose them all

$$\oint \vec{E} \cdot d\vec{a} = \sum_{i=1}^{n} \left(\oint \vec{E}_i \cdot d\vec{a} \right) = \sum_{i=1}^{n} \left(\frac{1}{\varepsilon_0} q_i \right)$$

For any closed surface

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\varepsilon_0} Q_{enc}$$ Gauss's law

Q_{enc} is the total charge enclosed within the surface

Gauss's law in integral form $$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\varepsilon_0} Q_{enc}$$

Use divergence theorem $$\oint_S \vec{E} \cdot d\vec{a} = \int_V \left(\vec{\nabla} \cdot \vec{E}\right) d\tau$$

$$Q_{enc} = \int_V \rho d\tau$$

Gauss' law will be $$\int_V \left(\vec{\nabla} \cdot \vec{E}\right) d\tau = \int_V \left(\frac{\rho}{\varepsilon_0}\right) d\tau$$

This should hold for any volume

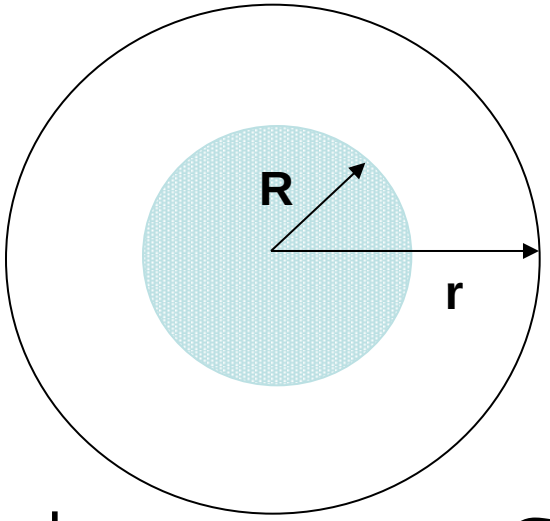
$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\varepsilon_0} \rho$$

"Gauss' law in differential form"

Applications of Gauss's law

Find the field outside a uniformly charged solid sphere of radius R and total charge q

Gaussian surface →



Consider a Gaussian spherical surface of radius r.

E is uniform at the surface of sphere in the direction $\mathbf{e}_r$

$$\int_S \left|\vec{E}\right| da = \left|\vec{E}\right| \int_S da = \left|\vec{E}\right| 4\pi r^2 = \frac{q}{\varepsilon_0}$$

or

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \hat{e}_r$$

Coulomb's law

Spherical symmetry — Make your Gaussian surface a concentric sphere

Cylindrical Symmetry — Gaussian surface a coaxial cylinder

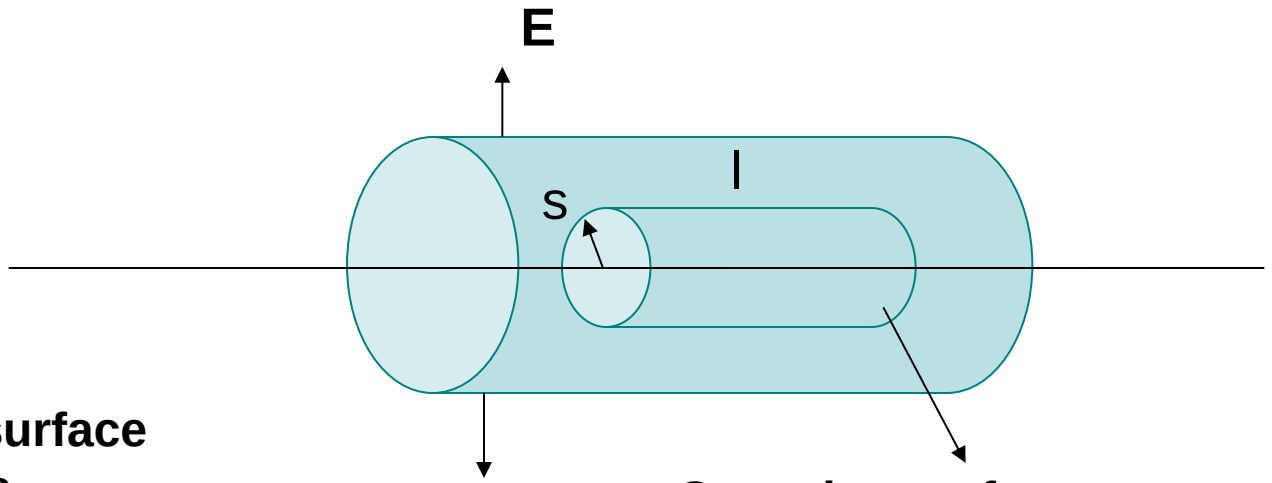
Plane symmetry — Gaussian surface a “pill box”

Gauss’s law is always true

But not always useful

Example.2

A long cylinder carries a charge density that is proportional to the distance from the axis: $\rho = ks$, for some constant K. Find the electric field inside the cylinder.



Consider a Gaussian surface of length l and radius s

Gaussian surface

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\varepsilon_0} Q_{enc}$$

Enclosed charge is

$$Q_{enc} = \int \rho d\tau = \int (ks')(s' ds' d\phi dz)$$

$$= 2\pi kl \int_0^s s'^2 ds' = \frac{2}{3}\pi kls^3$$

Symmetry dictates that E must point radially outward, so, for the curved portion of Gaussian cylinder

$$\int \vec{E} \cdot d\vec{a} = \int \left|\vec{E}\right| da = \left|\vec{E}\right| \int da = \left|\vec{E}\right| 2\pi s l$$

By Gauss's law

$$\left|\vec{E}\right| 2\pi s l = \frac{1}{\varepsilon_0} \frac{2}{3} \pi k l s^3$$

$$\vec{E} = \frac{1}{3\varepsilon_0} k s^2 \hat{s}$$